

MARTINGALES: WHAT HAPPENS WHEN WE STOP

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TO INFINITY AND BEYOND, WELL OR THE STOPPING TIME

What happens when we choose to stop observing a stochastic process? For example, a gambler to stop playing a game upon reaching a target amount, or an experiment ending once a particular event occurs. Such situations are modelled using stopping times, which gives us the chance to make decisions based solely on information available up until the present.

STOPPING TIMES

The concept of a stopping time helps us stop a process when something happens, *i.e.*, the process as it has been seen so far satisfies a certain property for the first time. Let $\{\mathcal{F}_n\}_{n \geq 0}$ be a filtration. For an experiment with a set of IID random variables, $X_1, X_2, \dots$, a stopping rule for $\{X_i\}_{i \geq 1}$ is a positive integer value random variable T (stopping time) such that for each $n \geq 1$, the event $\{T \leq n\} \in \mathcal{F}_n$ is independent of $X_{n+1}, X_{n+2}, \dots$, *i.e.*, to stop at n depends only on $X_1, X_2, \dots, X_n$. A martingale stopped at a stopping time is a martingale process.

2.1 Doob's Optional Stopping Theorem

We can use theorems, such as Doob's optional stopping theorem or Wald's equality, to analyse what happens when we decide to stop a random process. It won't tell us when to stop, but instead calculate the expected outcomes or properties of a process if you use a valid stopping rule (stopping time). Let $\{X_n\}_{n \geq 0}$ be a martingale w.r.t. itself and let T be a stopping time w.r.t. $\{X_n\}_{n \geq 0}$. Doob's optional stopping theorem states that if any of the following holds:

- (i) T is bounded ($\exists N < \infty$ s.t. $T(\omega) < N \forall \omega \in \Omega$),
- (ii) X_n is uniformly bounded ($|X_n(\omega)| < K \forall \omega \in \Omega$) and $\mathbb{P}(T = \infty) = 0$,
- (iii) $\mathbb{E}[T] < \infty$ and for some $K > 0$, $|X_n(\omega) - X_{n-1}(\omega)| \leq K \forall n \geq 1$ and $\omega \in \Omega$,

then $\mathbb{E}[X_T] = \mathbb{E}[X_0]$.

2.2 Wald's Equality

Let T be a stopping rule for IID random variables, $X_1, X_2, \dots$, and let $S_T = X_1 + X_2 + \dots + X_T$, if $\mathbb{E}[T] < \infty$ and $\mathbb{E}[|X|] < \infty$, then $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X]$.

Proof: Let $\mathbf{1}_n$ be the indicator function of the event, $\{T \geq n\}$. Therefore, $S_T = \sum_{n=1}^{\infty} X_n \cdot \mathbf{1}_n$, where:

$$\mathbf{1}_n = \begin{cases} 1 & \text{if } T \geq n \\ 0 & \text{if } T \leq n-1 \end{cases}$$

We know that $\mathbf{1}_n$ is independent of $X_n, X_{n+1}, \dots$, that is, it is dependent on $X_1, X_2, \dots, X_{n-1}$. Thus, $\mathbb{E}[X_n \cdot \mathbf{1}_n] = \mathbb{E}[X_n]\mathbb{E}[\mathbf{1}_n] = \mathbb{E}[X]\mathbb{E}[\mathbf{1}_n]$, and $\mathbb{E}[S_T] = \mathbb{E}\left[\sum_{n=1}^{\infty} X_n \cdot \mathbf{1}_n\right] = \sum_{n=1}^{\infty} \mathbb{E}[X_n \cdot \mathbf{1}_n] = \sum_{n=1}^{\infty} \mathbb{E}[X]\mathbb{E}[\mathbf{1}_n] = \mathbb{E}[X] \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_n] = \mathbb{E}[X]\mathbb{E}[T]$.

APPLICATIONS OF THE THEOREMS

One classical application of Doob's optional stopping theorem is to the gambler's ruin problem that we previously introduced in Article No.27. For example, we can use this theorem to find the probability that a fair random walk hits an upper boundary before a lower boundary. Consider a gambler who starts with $\pounds m$ and repeatedly plays a fair game. At each round, the gambler wins $\pounds 1m$ with probability $\frac{1}{2}$, and loses $\pounds 1m$ with probability $\frac{1}{2}$. The game continues until the gambler either reaches $\pounds Nm$ or is "ruined" by reaching zero. Let X_n denote the gambler's fortune after n rounds. Thus, $\{X_n\}_{n \geq 0}$ is a symmetric random walk and a martingale with respect to its natural filtration. What is the probability of success?

Solution: Let the stopping time, $T = \inf\{n \geq 0 : X_n = 0 \text{ or } X_n = N\}$. That is, T is the first time the gambler reaches either absorbing state, 0 or N . Let $p = \mathbb{P}(X_T = N \mid X_0 = i)$ be the probability that the gambler reaches $\pounds Nm$ before being ruined. Since $X_T \in \{0, N\}$, we have $\mathbb{E}[X_T] = 0 \cdot (1 - p) + N \cdot p = Np$. By Doob's Optional

Stopping Theorem, $\mathbb{E}[X_T] = \mathbb{E}[X_0] = i$. Thus, $Np = i \Rightarrow p = \frac{i}{N}$. Thus, for a fair game, the probability that a gambler starting with $\pounds i$ reaches $\pounds N$ before being ruined is $\mathbb{P}(X_T = N \mid X_0 = i) = \frac{i}{N}$. For example, if the gambler starts with $\pounds 2$ and aims to reach $\pounds 5$, then the probability of success is $\frac{2}{5}$.

We can apply Wald's equality to a plethora of questions with clear stopping rules. Suppose that you roll a dice and for each roll, you are paid the face value. If a roll lands on 4, 5, or 6, then you roll the dice again, and if a roll lands on 1, 2, or 3, then the game stops. What is the expected payoff of this game?

Solution: Do note that this could be solved via the law of total expectation, but, via Wald's equality, it can be solved very quickly. For each roll, the process has $\frac{1}{2}$ probability of stopping. Thus, the stopping time, T , follows a geometric distribution with $p = \frac{1}{2}$. We have $\mathbb{E}[T] = \frac{1}{p} = 2$. For each roll the expected face value is $\mathbb{E}[X] = \frac{7}{2}$. Thus, the expected payoff is $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X] = (2) \cdot \left(\frac{7}{2}\right) = 7$.

CONCLUSION

Stopping times can give us the chance to decide when to stop observing a stochastic process without relying on future information. Doob's optional stopping theorem and Wald's equality can allow expectations of complex random processes to be computed with simplicity.

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